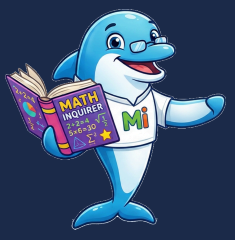
TUTOR GUIDE • LEVEL 1 • TERM 4 • LESSON 1 OF 10

Half and Quarter



Big idea: A half is one of 2 equal parts and a quarter is one of 4 equal parts, whatever the shape. Fold a shape so that the parts match. 2 equal parts are halves. 4 equal parts are quarters.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	2D Shapes: half circle and quarter circle (Primary 1) (Half circle and quarter circle as shapes, and equal parts of a shape (Primary 1, 2D Shapes 1.1, p. 32); Fractions as numbers, with notation such as 1/2 and 1/4, are Primary 2 (p. 33): this lesson uses the words half and quarter and pictures only, no fraction notation; Equal area in different shapes (the same half cut in different ways) and a fraction of a number of squares are Primary 2 and Primary 3 ideas (pp. 33 and 36)) Strand label for the tutor only. Start here is school Primary 1 work: fold a circle into halves and quarters, say whether a coloured part is a half or a quarter, count the equal parts, name a half circle and a quarter circle, and draw a line that cuts a square into 2 or 4 equal parts. No Primary 1 school paper in the sample we studied has an equal-parts item, so the wording is marked unverified; the Primary 1 papers do name and count shapes, which is what the warm-up uses. Think harder cuts a square into halves in several ways (DIAG), counts the small squares in a half and a quarter of a rectangle, tests four statements, and tests four statements. BEYOND PRIMARY 1 ON PURPOSE, at Think harder and Go further only: Same Half, Different Shape (three halves of a 2 by 4 rectangle with different shapes and the same 4 squares), guessing a shape from clues (LOG), and finding the whole from a quarter (BACK). Do not call this lesson MOE-aligned. Owner decision, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item and a non-routine type. Contest tag (tutor only): equal parts and fold lines; equal area in different shapes.
Objectives	• Fold a paper circle so the edges meet, say the fold makes 2 equal parts, and call each part a half. • Fold again, say the 4 equal parts are quarters, and say that 2 halves or 4 quarters make 1 whole. • Check whether parts are equal by laying one on top of the other, and say that parts that are not equal are not halves. • Name a half circle and a quarter circle. • Think harder: cut a square into halves in different ways, count the small squares in a half and a quarter of a rectangle, and test statements about halves and quarters. • Go further: see that the same half can have different shapes, guess a shape from clues, and find the whole when you know a quarter.
Joining cold	**Remember warm-up covers:** Share 8 things between 2 plates so that both plates match (Term 1 Lesson 8 and Term 2 Lesson 7, taught again with 8 dots); Name a circle, a square, a rectangle and a triangle (the holiday course, taught again with pictures); Share 7 things between 2 plates: 1 is left over, so the plates cannot match (Term 2 Lesson 4, odd and even, taught again). Only counting to 8 and the names of four shapes. Nothing from an earlier lesson is needed: the warm-up shares 8 dots between 2 plates, names four shapes and tries to share 7 dots between 2 plates. The words equal, half, quarter and whole are taught in Explore and the Mini-lesson. Try not to miss this lesson: Build the Figure uses a square folded corner to corner into 2 triangles, and Mi's Challenge Quest at the end of the year has a gate on halves and quarters. A child who misses it watches the recording, folds a paper circle twice with the kit page and does workbook pages 1 and 2 before the next lesson. A child who joins at the start of Term 4 needs nothing from Term 3.

Child materials	• Your Term 4 book (Book 7): the fold-shapes page (page 6), torn out along the dashed line, with the 10 paper shapes cut out by a grown-up beforehand: 4 circles, 4 squares and 2 long rectangles. Nobody cuts during the lesson; children only fold • No printer or no cutting? A grown-up cuts 4 circles (trace round a cup), 4 squares and 2 long strips from plain paper • A ruler (the one you keep at home), a pencil, a rubber, a coloured pencil, and spare paper for the warm-up • 8 small things such as coins or buttons for the warm-up, or draw 8 dots on paper
Tutor materials	• The slide deck • Your own folded circle and square to show on camera, if you wish • A visible clock A ruler is used to draw straight lines (Start here S6 and Think harder T1). Nothing needs to be bought. The paper shapes and the 8 small things stay on the table: nothing goes in the mouth or ears, and they are kept away from younger brothers and sisters. No scissors, matches, toothpicks or anything sharp in the lesson: a grown-up cuts the shapes beforehand. In class we use 2 circles (Explore), 1 long rectangle (Explore round 2) and up to 3 squares (Think harder and the Investigation at home); the rest are spares.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Same Half, Different Shape at G1, G3: A 2 by 4 rectangle (8 small squares) is cut into 2 halves in three different ways (a long bar, a block, a bent shape): every coloured half has 4 small squares and the shapes differ (G1). Equal parts need not be the same shape. G3 finds the whole from a quarter (a strip of 3 small squares is a quarter, the whole has 12). Level note: equal area in different shapes is a Primary 2 and Primary 3 fraction and area idea (pp. 33 and 36); a fraction of a number of squares is Primary 2. Named puzzle: Same Half, Different Shape. T1 (cut a square into halves in different ways) is the non-routine DIAG item and G2 the LOG item. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: draw it: cut a square into 2 equal parts in several different ways (DIAG); logic: guess the shape from its clues (LOG); same half, different shape: equal area is not equal shape; work backwards: from a quarter find the whole (BACK). Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, with your own paper shapes and a ruler. Check: S1 yes, no, yes; S2 4 equal parts, quarter; S3 any one part of the square and of the circle; S4 2 and 4; S5 shape 2 and shape 3; S6 one line and two lines through the middle; T1 three different cuts, Yes; T2 4 and 2; T3 true, false, true, false; G1 4, 4, 4, then No; G2 half circle, triangle; G3 12 and 6; I1 2, then at least two ways to make 4 equal parts; E1 quarter; E2 2; E3 No.
- Tell families before Term 4 starts (the goodbye in Lesson 30 says so): your Term 4 book is in the house; tear out the fold-shapes page along the dashed line; a grown-up cuts out the 10 shapes (4 circles, 4 squares, 2 long rectangles) and keeps them in an envelope; a ruler and a coloured pencil. No printer: a grown-up cuts 4 circles (trace round a cup), 4 squares and 2 long strips from plain paper. Also tell them the start date of Term 4.
- Have the slides open. Have your own folded circle and square to show on camera if you wish. Say equal, half, quarter and whole the same way every time, and keep the sentence: there are two equal parts, so each part is a half. Do not write or say 1/2 or 1/4.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Greet each child by first name as they arrive. Say the recording notice, then start the recording; once it is running, avoid first names. Nobody holds paper or a folded shape up to the camera.
- Be plain with families: halves and quarters of a circle are school Primary 1 shapes work (2D Shapes 1.1). No Primary 1 school paper in the sample we studied has an equal-parts item, so the wording is marked unverified. Cutting the same half into different shapes (Same Half, Different Shape), finding a half or a quarter of a number of squares and finding the whole from a quarter go beyond Primary 1 on purpose, at Think harder and Go further only. Do not call the lesson MOE-aligned.
- Slides s-explore1, s-explore2, s-explore3, s-m3 and s-m5 keep the picture or the answer off the screen while you ask: it appears on the slide after. Read every question aloud, including the shape numbers in S5 and the pictures in S1: the shapes on the slides are small.

Safeguarding and recording (this lesson)

- The paper shapes and the 8 small things for the warm-up stay on the table: nothing goes in the mouth or ears, and they are kept away from younger brothers and sisters. No scissors, matches, toothpicks or anything sharp. Nobody cuts during the lesson; a grown-up cuts the shapes beforehand and children only fold.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a folded shape up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. The turns are slow: a half turn and a quarter turn, and one whole turn done slowly; a child who feels dizzy sits down, and any child may stay seated and turn their hands or head.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.
- The gallery shows faces and fingers, not tables or papers. Ask for a finger answer when you need to see something. Answers on three are said quietly with the microphone off; a child who is invited to share switches the microphone on, then off again. The still-stuck photo is of the whole of workbook page 5 (Investigation, exit ticket and look-back).

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills

0–4	**Opening**	4	2: Half and Quarter, How our lessons work	3:12
4–10	**Remember warm-up**	6	3: Warm-up: share 8, Name each shape, Warm-up: share 7	4:56
10–14	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:14
14–26	**Explore**	12	6: Explore: fold a circle, Explore: fold a circle, Explore: fold again, Explore: fold again, Explore: a crooked fold, Explore: a crooked fold	10:06
26–36	**Mini-lesson**	10	7: A half, A quarter, Is half coloured?, Is half coloured?, Half circle and quarter circle, Your turn, Your turn	8:13
36–40	**Brain break**	4	1: Turn like a clock	3:14
40–70	**Workbook**	30	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	21:27
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:26
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:32
82–85	**Look-back**	3	1: Look-back	2:23
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:26
	Total	**90**		**66:12**

The script needs 66:12 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 78:13. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 14 (end of sneak-peek), minute 40 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 40, skip in this order: Start with: shorten Explore round 3 (the crooked fold) to a quick look and carry the idea to Talk time; Then: cut the second item of the Your turn slide (s-m5); Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 4:00 Opening

script 3:12 of 4:00

0:00	SLIDE	Show slide **Half and Quarter** (s-title)
0:08	DO	Greet each child by first name as they arrive, before the recording starts. Welcome anyone who is new this term: every idea today is taught from the start, and nobody needs Term 3. [0:25]
0:33	SAY	Hello, inquirers! Welcome to Term 4. Pencil ready, and your paper shapes on the table: the circles, the squares and the two long rectangles. Cameras can be on or off, as you like.
0:50	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
1:07	DO	Start the recording now. Do not use children's first names on the recording. [0:05]
1:12	REC	*If you are watching the recording: hello! Get a pencil, a ruler, paper and your paper shapes. Pause or rewind any time.*
1:23	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. If I invite you to talk to everyone, switch your microphone on, then off again. Nobody holds paper up to the camera.
1:46	SLIDE	Show slide **How our lessons work** (s-routine)
1:54	SAY	Four things to know. Listen first, then try: a guess is a good start. Answer on three. Tick your own work on the answer slide. If you are still stuck, tick the box on your look-back page and show your grown-up.
2:14	SAY	Today we fold paper into equal parts.
2:18	ASK	Think of a cake or a sandwich shared between two people. What does fair mean? [wait about 20 s for answers]
2:45	CHILDREN	Children whisper to a grown-up or to themselves what a fair share means. [0:25]
3:10	SAY	Let us warm up.
	NOTE	New this term? The routine slide takes about one minute and covers what a child who missed Terms 1 to 3 needs. Say the recording notice before you start the recording, and do not use children's first names once it is running. Say the recording statement and the microphone routine in the same words every lesson. Do not read out a child's name with an answer. The gallery shows faces and fingers, not tables: ask for a finger answer when you need to see something. The still-stuck route is the tick on the look-back page: a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the grown-up note, never on a slide).
	NOTE	Plain statement for families when it comes up: halves and quarters of a circle are school Primary 1 shapes work. Cutting the same half into different shapes, finding a half or a quarter of a number of squares and finding the whole from a quarter go beyond Primary 1 on purpose, at Think harder and Go further only. The paper shapes are folded, never cut, in the lesson.

4:00 – 10:00 Remember warm-up

script 4:56 of 6:00

4:00	SLIDE	Show slide **Warm-up: share 8** (s-rem1)
4:25	REC	*Recording: join in! Share the dots, name the shapes, and say your answers out loud before I do.*
4:34	SAY	Share eight dots between two plates. Draw eight dots on your paper, or lay out eight coins. Give one dot to plate A, one dot to plate B, and keep going until none are left. Both plates must match.
4:54	CHILDREN	Children share 8 dots between 2 plates one at a time. [0:35]

5:28	**ASK**	How many dots are on each plate? [wait about 12 s for answers]
5:44	**ASK**	How do you know both plates match? [wait about 18 s for answers]
6:06	**SLIDE**	Show slide **Name each shape** (s-rem2)
6:30	**SAY**	Now shapes. Look at shapes A, B, C and D.
6:36	**ASK**	What is the name of each shape? [wait about 20 s for answers]
6:59	**ASK**	How many straight sides has each one? [wait about 20 s for answers]
7:22	**SLIDE**	Show slide **Warm-up: share 7** (s-rem3)
7:48	**SAY**	Now share seven dots between the two plates, one at a time.
7:54	**CHILDREN**	Children share 7 dots between 2 plates one at a time and see 1 left over. [0:30]
8:24	**ASK**	Can both plates have the same? How do you know? [wait about 20 s for answers]
8:48	**SAY**	When the plates match, we say the parts are equal. Equal is our word for today.
	IF STUCK	If a child cannot share, give one dot to each plate in turn and count the plates. If a child is unsure of a shape name, point to the shape and say its name. For sides, touch each side with a finger.
	NOTE	Answers: 4 on each plate (both plates match); circle, square, rectangle, triangle; the circle has no straight side, the square and the rectangle 4, the triangle 3; no, 7 cannot be shared so that both match, 1 is left over. Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 8 and the shape names.

10:00 – 14:00 Sneak-peek puzzles

script 3:14 of 4:00

10:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
10:25	**REC**	*Recording: pause now and try both puzzles with a sheet of paper. Guessing is fine. Press play for the next part.*
10:36	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked. Use a sheet of scrap paper, not your kit shapes.
10:53	**READ**	Puzzle one. Fold a sheet of paper so the edges meet. Open it. How many parts has the fold made? Write the number.
11:06	**CHILDREN**	Children fold a sheet of paper and write the number on their own paper. [0:35]
11:41	**READ**	Puzzle two. Fold a sheet of paper so the edges meet. Fold it the same way again. Open it. How many equal parts are there? I know because blank.
11:56	**CHILDREN**	Children fold, open and write or draw their answer. [1:10]
13:06	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 2. Puzzle 2 answer: 4 equal parts. Look for good ideas to praise later (a child who said each fold cuts every part in 2, or who said 2 and 2 make 4). Do not write scores or keep a list.

14:00 – 26:00 Explore

script 10:06 of 12:00

14:00	**SLIDE**	Show slide **Explore: fold a circle** (s-explore1)
14:08	**REC**	*Recording: pause now. Fold your circle so the edges meet, open it, and lay one part on the other.*
14:18	**SAY**	Today your paper is the teacher. Take one paper circle from your kit page. Fold it so the edges meet exactly. Open it, and trace the fold with your pencil.
14:32	**CHILDREN**	Children fold a circle so the edges meet, open it and trace the fold. [1:10]
15:42	**SAY**	Now lay one part on top of the other.
15:47	**CHILDREN**	Children lay one part on the other and look. [0:20]
16:07	**ASK**	How many parts are there? Do they match? [wait about 15 s for answers]

16:26	**ASK**	How do you know? [wait about 20 s for answers]
16:48	**SLIDE**	Show slide **Explore: fold a circle** (s-explore1-ans)
16:56	**SAY**	Two parts that match are equal parts. Each part is a half. Two halves make one whole circle.
17:05	**SLIDE**	Show slide **Explore: fold again** (s-explore2)
17:13	**SAY**	Round two. Fold your circle in half again, so the new fold goes across the first fold. Open it and trace the new fold.
17:25	**CHILDREN**	Children fold the circle again, open it and trace the new fold. [1:10]
18:35	**ASK**	How many parts are there now? Do they all match? [wait about 20 s for answers]
19:00	**SAY**	Lay two of the parts on one half of your circle.
19:06	**CHILDREN**	Children lay 2 quarters on 1 half. [0:25]
19:30	**ASK**	What do you see? [wait about 12 s for answers]
19:44	**SAY**	Now take a long rectangle. It has three faint lines. Fold it in half, and in half again, so the folds land on the faint lines. Open it.
19:58	**CHILDREN**	Children fold the long rectangle twice and open it. [1:00]
20:58	**ASK**	How many strips are there? Do the folds land on the faint lines? [wait about 20 s for answers]
21:25	**SLIDE**	Show slide **Explore: fold again** (s-explore2-ans)
21:33	**SAY**	Four equal parts. Each part is a quarter. Four quarters make one whole. Two quarters make one half.
21:42	**SLIDE**	Show slide **Explore: a crooked fold** (s-explore3)
21:50	**SAY**	Round three. Take your second circle. This time fold it so the edges do not meet. Open it and trace the fold. Lay one part on the other.
22:04	**CHILDREN**	Children fold a circle crookedly, open it and lay one part on the other. [1:10]
23:14	**ASK**	Do the two parts match? [wait about 12 s for answers]
23:28	**ASK**	Are these two parts halves? How do you know? [wait about 20 s for answers]
23:53	**SLIDE**	Show slide **Explore: a crooked fold** (s-explore3-ans)
24:01	**SAY**	Two parts are not always halves. Halves have to be equal.
	IF STUCK	A child whose edges do not meet exactly folds again slowly: edge to edge, then press the fold flat. A child with no kit circle traces a cup. A child with no rectangle folds a long strip of paper twice. A child who lays parts on top and says they match has found the check for equal parts.
	NOTE	Main room only; children say what they have found. Tilting the camera to the table is allowed; a face need not be in view; nothing is held up to the camera. Answers: 2 parts, they match, a half; 4 parts, all match, a quarter, 2 quarters cover 1 half; 4 equal strips, the folds land on the faint lines; the crooked fold gives 2 parts that do not match, so not halves. Do not say the rule before the children do. Do not name a child with a wrong answer.

26:00 – 36:00 Mini-lesson

script 8:13 of 10:00

26:00	**SLIDE**	Show slide **A half** (s-m1)
26:08	**REC**	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
26:14	**SAY**	Here is a half. A half is one of two equal parts. Two halves make one whole. Look at the circle on the slide: the coloured part and the white part are the same size.
26:32	**CHILDREN**	Children fold their circle in half, open it and put a finger on one half. [0:30]
27:02	**ASK**	What makes a part a half? [wait about 18 s for answers]
27:23	**SLIDE**	Show slide **A quarter** (s-m2)

27:31	**SAY**	A quarter is one of four equal parts. Count the parts on this square with me. Four quarters make one whole.
27:42	**CHILDREN**	Children fold their circle in half twice, open it and put a finger on one quarter. [0:30]
28:12	**ASK**	How is a quarter different from a half? [wait about 18 s for answers]
28:34	**SLIDE**	Show slide **Is half coloured?** (s-m3)
28:42	**SAY**	Is half coloured? We can count. Each picture has six small squares. Count the coloured squares and the white squares.
28:52	**ASK**	In picture one, how many are coloured and how many are white? [wait about 15 s for answers]
29:12	**ASK**	In picture two, how many are coloured and how many are white? [wait about 15 s for answers]
29:34	**ASK**	In which picture is half coloured? [wait about 12 s for answers]
29:48	**SLIDE**	Show slide **Is half coloured?** (s-m3-ans)
29:56	**SAY**	Half means the two parts are the same size: three and three.
30:02	**ASK**	Without counting, how could we check that two parts are the same size? [wait about 22 s for answers]
30:31	**SAY**	We lay one part on top of the other. If they match, they are equal.
30:38	**SLIDE**	Show slide **Half circle and quarter circle** (s-m4)
30:46	**SAY**	Two new names. A half circle is half of a circle. A quarter circle is a quarter of a circle. The half circle has one straight side and one curved side.
31:02	**CHILDREN**	Children fold a circle in half and look at the folded shape. Then they fold it again and look again. [0:40]
31:42	**ASK**	What shape did you make the first time? And the second time? [wait about 20 s for answers]
32:08	**ASK**	How many half circles make a whole circle? [wait about 12 s for answers]
32:24	**ASK**	How many quarter circles make a whole circle? [wait about 12 s for answers]
32:40	**SLIDE**	Show slide **Your turn** (s-m5)
32:48	**SAY**	Your turn. Item one: is the coloured part a half or a quarter? Item two: which is bigger, a half or a quarter of the same circle? Say it on three, or write it on your paper.
33:06	**CHILDREN**	Children write or say the two answers. Wait 10 seconds before helping. [0:50]
33:56	**SLIDE**	Show slide **Your turn** (s-m5-ans)
34:04	**READ**	Item one: a quarter. Item two: a half. Tick your own work with a tick.
	NOTE	Key words to use every time: equal, half, quarter, whole, parts. Say it the same way each time: there are two equal parts, so each part is a half. Do not use the word wrong for a child, only for a picture or a statement. No fraction notation is written or spoken.
	IF STUCK	If a child cannot tell a half from a quarter, ask them to count the parts: two parts, half; four parts, quarter. If a child is unsure whether parts are equal, lay one on the other, or count the small squares.

36:00 – 40:00 Brain break

script 3:14 of 4:00

36:00	**SLIDE**	Show slide **Turn like a clock** (s-break)
36:08	**REC**	*Recording: stand up and join in! Make a whole turn, a half turn and a quarter turn, slowly.*
36:17	**SAY**	Stand up and get a sip of water. Face the screen. We will turn slowly. A half turn: turn until you face the back. Now turn back. A quarter turn: turn until you face the side. Turn back.
36:36	**MOVE**	Call a half turn and a quarter turn slowly, turning back each time. Then one whole turn, very slowly, all the way round. A child who sits turns their hands or head.
39:06	**SAY**	A whole turn is two half turns. It is also four quarter turns. Sit down and breathe.

40:00 – 70:00 Workbook

script 21:27 of 30:00

40:00	SLIDE	Show slide **Workbook page 1: Start here** (s-ws1)
40:08	REC	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
40:17	READ	Page 1, Start here. Question 1: look at the three shapes. Is half of each shape coloured? Circle Yes or No for each. Question 2: look at shape B in question 1. Fill in how many equal parts, then circle half or quarter. Question 3: colour one half of the square, then colour one quarter of the circle.
40:49	CHILDREN	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [6:40]
47:29	SLIDE	Show slide **Check page 1** (s-ws1ans)
47:37	READ	Question 1: shape A yes, shape B no, shape C yes. Question 2: 4 equal parts; a quarter. Question 3: any one part of the square and any one part of the circle. Tick your own page with a tick.
47:58	SAY	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
48:07	SLIDE	Show slide **Workbook page 2: Start here** (s-ws2)
48:15	READ	Page 2. Question 4: fill in how many halves make a whole circle, and how many quarters make a whole square. Question 5: look at the four shapes. Write the number of the half circle and the number of the quarter circle. Question 6: use your ruler. Draw one line to cut square A into 2 equal parts. Draw two lines to cut square B into 4 equal parts.
48:53	CHILDREN	Page 2, questions 4 to 6. [7:00]
55:53	SLIDE	Show slide **Check page 2** (s-ws2ans)
56:01	READ	Question 4: 2 halves; 4 quarters. Question 5: the half circle is shape 2; the quarter circle is shape 3. Question 6: one line through the middle; two lines through the middle that cross. Tick your own page with a tick.
56:23	SLIDE	Show slide **Choose your next page** (s-choice)
56:31	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
56:47	MOVE	Stand and reach up for 30 seconds, then sit down and choose.
57:17	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [4:10]
	IF STUCK	Q1: count the parts, then check the coloured part and the white part match. Q2: look back at shape B and count its parts. Q3: the shape is already cut into parts: colour one part. Q4: think of the circle cut into halves and the square cut into quarters. Q5: the half circle is half of a circle, the quarter circle is a quarter of a circle. Q6: fold a paper square to find where the line goes.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder cuts a square into halves in different ways, counts squares in a half and a quarter, and tests four statements. Go further has Same Half, Different Shape, two shape riddles and a quarter that gives the whole (beyond Primary 1 on purpose). Do not announce who is on which page. Start here takes a middle child about 17 minutes, so the choice time is short: for most children pages 3 to 5 are for home. Say so plainly, do not skip the answer slides, and take any extra minutes from the flex time.

70:00 – 76:00 Talk time

script 4:26 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up whether any 2 parts of a shape are halves, and test it with your circle.*

70:35	**READ**	A child says: any two parts of a shape are halves. Is the child right? Test it with your circle.
70:46	**CHILDREN**	Test it with your circle: fold it crookedly, or look at your crooked fold from Explore. [0:40]
71:26	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:06	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 80 s for answers]
73:32	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
73:57	**SAY**	Remember the sneak-peek? Puzzle one: two parts. Puzzle two: four equal parts. The second fold cuts each of the two parts in two.
74:08	**ASK**	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: no. Halves must be equal, so they must match when one is laid on the other. A crooked fold makes two parts that do not match. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:32 of 6:00

76:00	**SLIDE**	Show slide **Exit ticket** (s-exit)
76:25	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	**READ**	E1. A circle is cut into 4 equal parts. Is each part a half or a quarter? Circle one. E2. Fill in the blank: blank halves make 1 whole. E3. A square is cut into 2 parts that are not the same size. Are they halves? Circle Yes or No.
77:08	**CHILDREN**	Children answer E1 to E3 on their own. [2:50]
79:58	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:23	**READ**	E1: a quarter. E2: 2. E3: No.
80:27	**SAY**	Tick your own ticket with a tick. Well done.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:23 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
83:03	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:26 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:08	**SAY**	Well done, inquirers. Next lesson is Build the Figure. Please have the shape pieces page cut out by a grown-up, twenty paper pieces, kept in an envelope, and a pencil. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Optional: corner to corner** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: Mei folds a paper square corner to corner. Each part is a half. What shape is each part?

CHILDREN	Short think time, then take two answers. Children check by folding a spare square. [1:30]
SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: a triangle. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content. If you are more than 3 minutes ahead at minute 40, use one menu activity then.
- A few children ahead: a few children may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask "How do you know?" and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	A child says: a quarter is always smaller than a half. Is that always true? Think about a quarter of a big sheet of paper and a half of a small card.	*Recording: pause. Is a quarter always smaller than a half? Think of a quarter of a big sheet and a half of a small card.*	Not always. A quarter and a half of the same shape: the quarter is smaller. But a quarter of a big sheet can be bigger than a half of a small card. We can only compare when the whole is the same.
Make a puzzle for your grown-up	4 (optional)	Fold a paper square in a secret way. Open it so only the folds show. Ask a grown-up: how many equal parts? Is each part a half or a quarter? Check by laying the parts on top of each other.	*Recording: pause. Fold a square in a secret way. Ask a grown-up: how many equal parts, and half or quarter?*	No single answer. A good puzzle has parts that really match: 2 equal parts are halves, 4 equal parts are quarters.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never met halves. Show how to check whether two parts are halves: lay one part on top of the other.	*Recording: pause. Teach a grown-up how to check whether two parts are halves, by laying one on top of the other.*	Two parts are halves if there are exactly 2 of them and they match when one is laid on the other.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 10 to 14)

Puzzle 1 (access): Fold a sheet of paper so the edges meet. Open it. How many parts has the fold made? Write the number.

> **Answer: 2**

Puzzle 2 (think): Fold a sheet of paper so the edges meet. Fold it the same way again. Open it. How many equal parts are there? I know because ___.

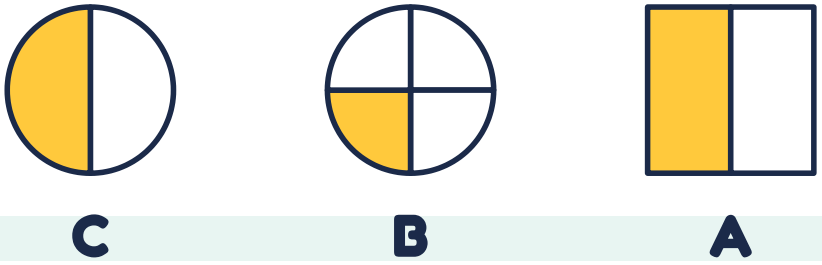
> **Answer: 4. The first fold makes 2 equal parts. The second fold cuts each of those parts into 2 equal parts, so 2 and 2 make 4.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Look at the three shapes. Is half of each shape coloured? Circle Yes or No for each shape.**

(a) Shape A. Circle Yes or No.
(b) Shape B. Circle Yes or No.
(c) Shape C. Circle Yes or No.



> **Answer: A yes; B no; C yes**

Hint (say this first): A half means 2 equal parts. Count the parts of each shape. Is one of 2 parts coloured?
Worked solution: Shape A has 2 equal parts and 1 is coloured, so half is coloured: Yes. Shape B has 4 equal parts and 1 is coloured, so a quarter is coloured, not a half: No. Shape C has 2 equal parts and 1 is coloured, so half is coloured: Yes.
Common mistakes: Saying Yes for any coloured part.; Not counting the parts: a quarter is coloured in shape B.
Time for a middle child: about 3 min.

S2 **Look at shape B in question 1. Fill in the blanks.**

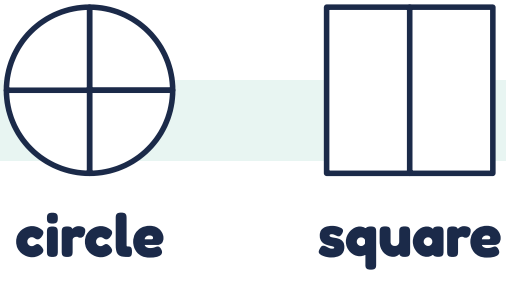
(a) Shape B has ☐ equal parts.
(b) Each part is a half or a quarter. Circle one.

> **Answer: 4 equal parts; quarter**

Hint (say this first): Touch each part of shape B and count. Then think: is a part one of 2 equal parts or one of 4 equal parts?
Worked solution: Shape B has 4 equal parts. One of 4 equal parts is a quarter.
Common mistakes: Counting the coloured part only and writing 1.; Writing half: a half is one of 2 equal parts.
Time for a middle child: about 2 min.

S3 **Colour the shapes.**

(a) Colour one half of the square.
(b) Colour one quarter of the circle.



> **Answer: any one part of the square; any one part of the circle**

Hint (say this first): The shapes are already cut into equal parts. A half is 1 of the 2 parts. A quarter is 1 of the 4 parts.
Worked solution: The square has 2 equal parts: colour 1 of them for a half. The circle has 4 equal parts: colour 1 of them for a quarter.
Common mistakes: Colouring 2 parts of the circle.; Colouring both parts of the square.
Time for a middle child: about 2.5 min.

S4 **Fill in the blanks.**

(a) ☐ halves make 1 whole circle.

(b) ☐ quarters make 1 whole square.

Answer: 2; 4

Hint (say this first): Think of a circle cut into halves and a square cut into quarters. All the parts together make 1 whole.

Worked solution: A circle cut into 2 equal parts has 2 halves, and 2 halves make 1 whole. A square cut into 4 equal parts has 4 quarters, and 4 quarters make 1 whole.

Common mistakes: Writing 1, the number of wholes.; Mixing up halves and quarters.

Time for a middle child: about 2 min.

S5 **Look at the four shapes. Write the number of the shape.**



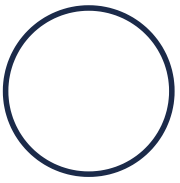
4



3



2



1

(a) The half circle is shape ☐.

(b) The quarter circle is shape ☐.

Answer: 2; 3

Hint (say this first): A half circle is half of a circle. A quarter circle is a quarter of a circle: it has a corner.

Worked solution: Shape 2 is a half circle: half of a circle, with 1 straight side. Shape 3 is a quarter circle: a quarter of a circle, with 2 straight sides that make a corner.

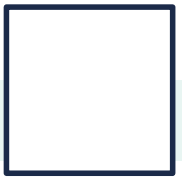
Common mistakes: Choosing the whole circle for the half circle.; Mixing up shapes 2 and 3.

Time for a middle child: about 3 min.

S6 **Use your ruler. Draw straight lines on the squares.**

(a) Square A. Draw 1 line to make 2 equal parts.

(b) Square B. Draw 2 lines to make 4 equal parts.



B



A

Answer: square A: 1 line through the middle; square B: 2 lines through the middle that cross

Hint (say this first): Fold a paper square to find where the line goes. The parts must match.

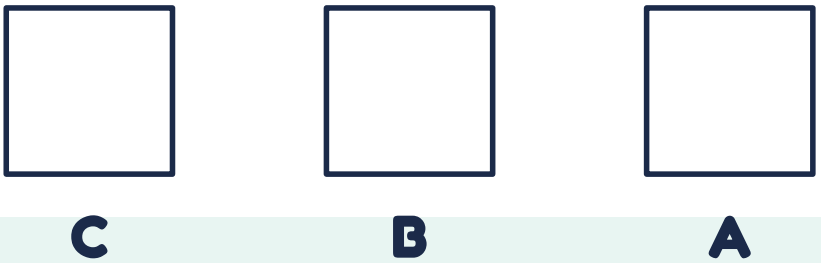
Worked solution: For square A, 1 line through the middle (across, down or corner to corner) makes 2 equal parts. For square B, 2 lines through the middle that cross make 4 equal parts.

Common mistakes: Drawing a line that is not through the middle, so the parts do not match.; Drawing 4 lines instead of 2.

Time for a middle child: about 3.5 min.

Think harder (page 3): the taught idea in a new setting

T1 **Cut each square into 2 equal parts. Use your ruler. Make a different cut each time.**

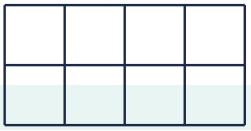


(a) Fold a paper square on one of your lines. Do the 2 parts match? Circle Yes or No.

Answer: any three different lines through the middle: down, across, corner to corner

Hint (say this first): Fold a paper square to try a cut first. If the 2 parts match when you lay one on the other, they are halves.
Worked solution: Down the middle, across the middle and corner to corner each make 2 equal parts. The parts match when we fold, so each part is a half. A half of a square can be a rectangle or a triangle.
Common mistakes: Drawing a line that is not through the middle.; Drawing the same cut on all three squares.
Time for a middle child: about 5 min.

T2 **The rectangle has 8 small squares. Fill in the blanks.**



(a) A half of the rectangle is ☐ small squares.
(b) A quarter of the rectangle is ☐ small squares.

Answer: 4; 2

Hint (say this first): A half is 2 equal parts, so share the 8 squares into 2 groups that match. A quarter is 4 equal parts.
Worked solution: A half is 2 equal parts of 8 squares: 4 and 4, so a half is 4 squares. A quarter is 4 equal parts: 2, 2, 2 and 2, so a quarter is 2 squares.
Common mistakes: Writing 8 for the half.; Writing 4 for the quarter.
Time for a middle child: about 3.5 min.

True/False statements: A half of a square can be a triangle.; Any 2 parts of a shape are halves.; 2 quarters of a circle make 1 half of the circle.; A half circle is the same size as a quarter circle.

T3 **Tick True or False for each statement.**

Answer: true; false; true; false

Hint (say this first): Try each statement with a paper circle or square. Fold and lay the parts on top of each other.
Worked solution: A half of a square can be a triangle: True (fold corner to corner). Any 2 parts of a shape are halves: False, they must be equal. 2 quarters of a circle make 1 half of the circle: True (we laid 2 quarters on 1 half). A half circle is the same size as a quarter circle: False, a half circle is bigger.
Common mistakes: Saying True for any 2 parts of a shape.; Thinking a half cannot be a triangle.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **Same Half, Different Shape. Each rectangle has 8 small squares. A half of each is coloured. Look at the three rectangles.**



(a) Count the coloured squares. Rectangle A: ☐ Rectangle B: ☐ Rectangle C: ☐
(b) Do the three coloured parts have the same shape? Circle Yes or No.

Answer: 4, 4, 4; No

Hint (say this first): Count the coloured squares in each rectangle. Then look at the shape of the coloured part: is it a long bar, a block or a bent shape?
Worked solution: Each coloured part has 4 small squares, and the white part has 4 as well, so each coloured part is a half. The coloured parts have different shapes: a long bar, a block of 4 and a bent shape. Equal parts do not have to be the same shape.
Common mistakes: Thinking halves must look the same.; Counting the white squares and writing 4 for only one rectangle.
Time for a middle child: about 6 min.

G2 **Guess the shape. Circle my name.**
(a) I have 1 straight side and 1 curved side. 2 of me make a circle.
(b) I am cut from a square. I have 3 corners. 2 of me make the square.

Answer: half circle; triangle

Hint (say this first): Think of the shapes from today. Which one has 1 straight side and 1 curved side? Which one has 3 corners?
Worked solution: Riddle a: 1 straight side and 1 curved side, and 2 of me make a circle: a half circle. Riddle b: cut from a square, 3 corners, and 2 of me make the square: a triangle.
Common mistakes: Choosing the quarter circle for riddle a: 4 of those make a circle, not 2.; Choosing the rectangle for riddle b.
Time for a middle child: about 5 min.

G3 **The strip is a quarter of a rectangle. Fill in the blanks.**
(a) The whole rectangle has ☐ small squares.
(b) A half of the rectangle has ☐ small squares.



Answer: 12; 6

Hint (say this first): 4 quarters make the whole. The strip is 1 quarter. How many small squares in 4 of them?
Worked solution: The strip has 3 small squares and it is 1 quarter. 4 quarters make the whole: 3, 6, 9, 12. The whole has 12 small squares. A half is 2 quarters: 6 small squares.
Common mistakes: Writing 4 or 7 (adding instead of making 4 equal groups).; Writing 12 for the half.
Time for a middle child: about 4.5 min.

Investigation (page 5): Fold twice

Easy first step: Fold a paper square in half. Open it. How many parts? Write the number.
Then: Fold it in half again to make 4 equal parts. Draw the fold lines on the first square. Make 4 equal parts a different way on the second square.

Model answer: First: 2 parts. Then at least 2 different ways, and 3 are easy to find: two lines down and across (4 small squares), two corner-to-corner lines (4 triangles), or 3 parallel lines (4 strips). Each way gives 4 equal parts, but the parts have different shapes. Fold the square to check that the parts match.

Look for: A child who folds a paper square to test a way, who says 4 equal parts can look different (squares, triangles, strips), and who checks by laying the parts on top of each other. Credit any clear reason. Do not mark a child down for stopping at one way.
Common mistakes: Drawing 3 lines that make parts that are not the same size.; Drawing the same way on both squares.

Talk and explain (minutes 70 to 76)

A child says: any 2 parts of a shape are halves. Is the child right? Test it with your circle. Sentence starter: I think … because …

> **Model answer: No. Halves must be equal, so they must match when one is laid on the other. A crooked fold makes 2 parts that do not match, so they are not halves.**

Look for: A child who tests it with a crooked fold, or says the parts must match or be the same size. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **A circle is cut into 4 equal parts. Is each part a half or a quarter? Circle one.**

> **Answer: quarter**

Worked solution: The circle has 4 equal parts, so each part is a quarter.
Common mistakes: Saying half for any part.; Not counting the parts.

E2 **Fill in the blank.**

☐ halves make 1 whole.

> **Answer: 2**

Worked solution: 2 equal parts make 1 whole, and each part is a half.
Common mistakes: Writing 1.; Writing 4, the quarters.

E3 **A square is cut into 2 parts that are not the same size. Are they halves? Circle Yes or No.**

> **Answer: No**

Worked solution: Halves must be equal. Parts that are not the same size are not halves.
Common mistakes: Saying Yes because there are 2 parts.; Thinking the bigger part is a half.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (folding the paper; knowing if parts are equal; half or quarter; something else); *What helped me* (my paper shapes; laying one part on the other; checking my answer; I was not stuck); *Next time I will try* (folding carefully; checking the parts match; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	A circle is cut into 4 equal parts. Each part is a ____ . Write half or quarter.	quarter
H2	Start here	Fill in the blank: ____ quarters make 1 whole.	4
H3	Think harder	A rectangle has 12 small squares. A half of it is ____ small squares.	6
H4	Go further	A big square has 16 small squares. A quarter of it is ____ small squares. A half of it is ____ small squares.	4 and 8
H5	Start here	Kumar is 7th in the line. There are ____ children in front of Kumar.	6 children

Online practice (`out/l31/practice`, key `t4l31`): 15 items; 6 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning that a whole can be cut into equal parts. This week: halves and quarters of a circle and a square.
Term line (plan, section 5): Your child is learning to choose a way to solve a problem, and to make problems of their own. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–4	**Half and Quarter** s-title	TODAY WE WILL: / Fold shapes into equal parts.	Lesson 1 of Term 4, 90 minutes. Objective: fold a circle into halves and quarters, check that parts are equal by laying one on the other, name a half circle and a quarter circle, and meet the beyond-Primary-1 puzzle Same Half, Different Shape. Plain statement for families: halves and quarters of a circle are Primary 1 shapes work; cutting the same half into different shapes, finding a half or a quarter of a number of squares and finding the whole from a quarter go past the school Primary 1 syllabus on purpose, at Think harder and Go further only. Time plan: opening 0 to 4, warm-up 4 to 10, sneak-peek 10 to 14, explore 14 to 26, mini-lesson 26 to 36, brain break 36 to 40, workbook 40 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice in one line: you and the screen only; a child may keep the camera off.
2	0–4	**How our lessons work** s-routine	Listen first, then try. A guess is a good start. / Answer on three: say it quietly, show fingers or a thumbs up. / Tick your own work on the answer slide. / Still stuck? Tick the box on your look-back page and show your grown-up.	New-here card, about one minute. Read the four lines slowly. It is the same card in every opening of the term and it covers what a child who joined this term needs. Nobody holds paper to the camera: answers are said quietly on three (microphones stay off), shown with fingers or a thumbs up, or ticked on the child's own page. The still-stuck tick is private: a grown-up may photograph the look-back page after class and send it to the business WhatsApp.
3	4–10	**Warm-up: share 8** s-rem1	Share 8 dots between 2 plates. Both plates must match. / [picture: plates] / How many dots go on each plate?	Children draw 8 dots on a scrap of paper (or lay out 8 coins or buttons) and share them between 2 plates, one at a time, until none are left. Both plates must match. Ask: how many on each plate, and how do you know the plates match. The slide shows empty plates and never the count. This is the idea of equal sharing that the lesson turns into halves. A child who is unsure shares one dot to each plate in turn. No writing, no marks.
4	4–10	**Name each shape** s-rem2	[picture: shapes2d] / [picture: shapes2d] / [picture: shapes2d] / [picture: shapes2d] / Say the name of each shape. How many sides has each one?	Four shapes labelled A to D: A circle, B square, C rectangle, D triangle. Children say each name aloud on three, then how many sides each has (the circle has none that are straight; the square 4; the rectangle 4; the triangle 3). This is the holiday-course naming, taught again with pictures so a child who joins cold can follow. Ask: what is the same about the square and the rectangle (4 sides, 4 corners). The slide never shows the names. A child who is unsure touches each side with a finger.
5	4–10	**Warm-up: share 7** s-rem3	Share 7 dots between 2 plates. / [picture: plates] / Can both plates match?	Children try to share 7 dots between 2 plates, one at a time. One dot is left over, so the plates cannot match. Ask: can both plates have the same (no); what is left over (1 dot). Say: when the plates match we say the parts are equal. The word equal is the word for today. The answer is not on the slide. A child who counted 3 and 4 says that the plates are not the same, and that is the right idea.
6	10–14	**Sneak-peek puzzles** s-sneak	PUZZLE 1: anyone can try: / Fold a sheet of paper so the edges meet. Open it. How many parts has the fold made? / PUZZLE 2: think: / Fold a sheet of paper so the edges meet. Fold it the same way again. Open it. How many equal parts are there? I know because ___.	Both puzzles use a sheet of scrap paper (not the kit shapes). Puzzle 1 is an access question: any child can fold a sheet and count 2 parts. Puzzle 2 is a thinking question on the lesson idea: after two folds there are 4 equal parts, because each fold cuts every part into 2. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 2; 4. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.

7	14–26	**Explore: fold a circle** s-explore1	Take one paper circle from your kit. / Fold it so the edges meet. / Open it. Trace the fold with a pencil. / Lay one part on top of the other. Do they match? / Fold only. Nobody cuts.	Round 1, about 4 minutes. Each child has a paper circle from the kit page (or a circle they traced round a cup). Fold so the edges meet exactly, open, trace the fold with a pencil, then lay one part on top of the other: they match, so the 2 parts are equal. Say the words: equal parts; each part is a half; 2 halves make 1 whole. Ask: how many parts; do they match; how do you know. The answer picture is on the next slide, after the children have answered. Answers: 2 parts, they match, a half. A child whose edges do not meet exactly tries again with a slower fold.
8	14–26	**Explore: fold a circle** s-explore1-ans	Take one paper circle from your kit. / Fold it so the edges meet. / Open it. Trace the fold with a pencil. / Lay one part on top of the other. Do they match? / Fold only. Nobody cuts. / [picture: shapes2d] / 2 EQUAL PARTS: / Each part is a half. / 2 halves make 1 whole.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
9	14–26	**Explore: fold again** s-explore2	Fold your circle in half again. / Open it. Trace the new fold. How many parts now? / Lay the parts on top of each other. Do they all match? / Now take a long rectangle. Fold it in half, then in half again. Open it.	Round 2, about 5 minutes. Fold the same circle in half again so the fold lands across the first one; open it: 4 parts that all match, so they are quarters. Ask a child to lay 2 quarters on 1 half: they cover it exactly, so 2 quarters make 1 half. Then the long rectangle from the kit: it has 3 faint lines. Fold it in half, then in half again so the folds land on the faint lines; open it. 4 equal strips, so each strip is a quarter of the rectangle. Answers: 4 parts; yes they match; a quarter; the folds land on the faint lines. A child with no rectangle folds a long strip of paper twice.
10	14–26	**Explore: fold again** s-explore2-ans	Fold your circle in half again. / Open it. Trace the new fold. How many parts now? / Lay the parts on top of each other. Do they all match? / Now take a long rectangle. Fold it in half, then in half again. Open it. / [picture: shapes2d] / 4 EQUAL PARTS: / Each part is a quarter. / 4 quarters make 1 whole. / 2 quarters make 1 half.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
11	14–26	**Explore: a crooked fold** s-explore3	Take your second circle. / Fold it so the edges do not meet. / Open it. Trace the fold. / Lay one part on the other. Do they match? / Two parts, but are they halves?	Round 3, about 3 minutes. Each child folds a second circle crookedly: the edges must not meet, so the fold does not go through the middle. Open it: 2 parts, one big and one small. Lay one on the other: they do not match. Say: two parts are not always halves. Halves must be equal. Ask: are these two parts halves (no); how do you know (they do not match). The answer is on the next slide. A child whose crooked fold turned out to be a good half is asked to fold again a different way, and the good half is praised as a good fold.
12	14–26	**Explore: a crooked fold** s-explore3-ans	Take your second circle. / Fold it so the edges do not meet. / Open it. Trace the fold. / Lay one part on the other. Do they match? / Two parts, but are they halves? / The parts do not match. They are not equal, so they are not halves.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.

13	26–36	**A half** s-m1	A HALF: / One of 2 equal parts. / 2 halves make 1 whole. / [picture: shapes2d]	Name the idea in words. A half is one of 2 equal parts. Point to the coloured part and the white part of the circle on the slide: they are the same size, so each is a half. Say: 2 halves make 1 whole circle. Children say it with you. Ask: what makes a part a half (there are 2 parts and they match). There is no fraction notation in this lesson: the word half only. Keep the picture on the screen while you speak.
14	26–36	**A quarter** s-m2	A QUARTER: / One of 4 equal parts. / 4 quarters make 1 whole. / [picture: shapes2d]	A quarter is one of 4 equal parts. The square on the slide is cut into 4 equal parts with one coloured. Children count the parts and say: 4 equal parts, so one part is a quarter. Say: 4 quarters make 1 whole square. Ask: how is a quarter different from a half (4 parts instead of 2, so each part is smaller). The word quarter will return when we read the clock in Term 3; today it means one of 4 equal parts.
15	26–36	**Is half coloured?** s-m3	Count the coloured squares and the white squares. / [picture: grid] / Picture 1 / [picture: grid] / Picture 2 / In which picture is half coloured?	A check that works without folding: count. Each picture is 6 small squares. Picture 1 has 3 coloured and 3 white, so the 2 parts match and half is coloured. Picture 2 has 4 coloured and 2 white, so they do not match and half is not coloured. Ask: how many coloured; how many white; do they match. The answer appears on the next slide, after the children have answered. Say: half means the two parts are the same size. Answers: Picture 1; 3 and 3; 4 and 2.
16	26–36	**Is half coloured?** s-m3-ans	Count the coloured squares and the white squares. / [picture: grid] / Picture 1 / [picture: grid] / Picture 2 / In which picture is half coloured? / Picture 1: 3 coloured and 3 white match, so half is coloured. Picture 2: 4 coloured and 2 white do not match.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
17	26–36	**Half circle and quarter circle** s-m4	[picture: shapes2d] / [picture: shapes2d] / [picture: shapes2d] / A half circle is half of a circle. A quarter circle is a quarter of a circle.	Two new shape names. The half circle has 1 straight side and 1 curved side. The quarter circle has 2 straight sides that meet at a corner and 1 curved side. Ask: how many half circles make a whole circle (2); how many quarter circles (4). Children can hold up their folded paper: a folded circle shows a half circle, and folded twice shows a quarter circle. Say the names together twice. This is the Primary 1 shape work in the lesson (2D Shapes 1.1).
18	26–36	**Your turn** s-m5	ITEM 1: / Is the coloured part a half or a quarter? / [picture: shapes2d] / ITEM 2: / Which is bigger, a half or a quarter of the same circle?	Two quick items, answered on the children's own paper or said quietly on three (say so aloud). Item 1: a rectangle in 4 equal parts with one coloured, so it is a quarter. Item 2: a half is bigger than a quarter of the same circle because a circle in 2 parts has bigger parts than a circle in 4 parts. Walk the cameras and wait 10 seconds before helping. Nobody holds paper up. The next slide shows the answers and children tick their own work. Answers: a quarter; a half.
19	26–36	**Your turn** s-m5-ans	ITEM 1: / Is the coloured part a half or a quarter? / [picture: shapes2d] / ITEM 2: / Which is bigger, a half or a quarter of the same circle? / Item 1: a quarter. The rectangle has 4 equal parts. Item 2: a half. 2 parts are bigger than 4 parts of the same circle.	Answer slide for the previous slide. Read the answer aloud after the children have answered, then give the idea in your own words. The notes on the previous slide have the answers and the common slips.
20	36–40	**Turn like a clock** s-break	Stand up. Face the screen. / A half turn. A quarter turn. / Sip some water.	Four minutes. Stand up with a little clear space, in socks or soft shoes, no running. Call the turns slowly: a half turn (you face the back), then turn back; a quarter turn (you face the side), then turn back; a whole turn once, very slowly, all the way round. Say: a whole turn is 4 quarter turns and 2 half turns. A child may stay seated and turn their hands or head. Finish with a sip of water. No equipment. Do not give prizes or compare children.

21	40–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own with their paper shapes; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 1: is half coloured, for three shapes. Question 2: shape B again, how many equal parts and what is each part called. Question 3: colour one half and one quarter.
22	40–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 yes, no, yes; Q2 4 equal parts, quarter; Q3 any one of the 2 parts of the square and any one of the 4 parts of the circle.
23	40–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Take your time. / questions S4, S5, S6	Read questions 4 to 6 aloud. Question 4: how many halves make a whole circle, how many quarters make a whole square. Question 5: find the half circle and the quarter circle among four shapes. Question 6: use a ruler to draw lines that cut a square into 2 equal parts and another into 4 equal parts. Children may fold a paper square to check. Wait 10 seconds before helping.
24	40–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q4 2 halves and 4 quarters; Q5 the half circle is shape 2 and the quarter circle is shape 3; Q6 one line through the middle makes 2 equal parts, two lines through the middle make 4 equal parts.
25	40–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Cut a square in different ways / Go further (page 4) | Same Half, Different Shape / Investigation (page 5) | Fold twice / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
26	70–76	**Talk and explain** s-talk	A CHILD SAYS: / Any 2 parts of a shape are halves. / Is the child right? Test it with your circle. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Children test it with their second circle from Explore round 3 (the crooked fold), or fold a fresh one crookedly. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. Halves must be equal, so they must match when one is laid on the other. A crooked fold makes 2 parts that do not match. Repeat each child answer in your own words for the recording. The child who is wrong is a child in another class: no cast name is given the wrong answer.
27	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: 2 parts. Puzzle 2: 4 equal parts. The second fold cuts each of the 2 parts into 2.	Reveal both puzzles now. Say: many of you wrote a good reason. Do not read out anyone earlier answer. Puzzle 1: one fold makes 2 parts. Puzzle 2: the second fold cuts both of the 2 parts in 2, so 2 and 2 make 4 equal parts, which are quarters. It was the same idea as Explore round 2.
28	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, and read the words Yes and No and half and quarter. Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
29	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: a quarter; 2; No. Do not read out any child result.

30	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
31	85–90	**Well done, inquirers!** s-end	Show a grown-up one shape you folded today. / NEXT LESSON: / Build the Figure / Have ready: the shape pieces page, cut out by a grown-up (20 pieces), a pencil and paper.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson needs the shape-pieces page from the Term 4 book, with the 20 pieces cut out by a grown-up and kept in an envelope, and a pencil. If the whole class is ahead, show the optional slide and use the Spare-time menu from the tutor guide.
32	optional	**Optional: corner to corner** s-opt	Mei folds a paper square corner to corner. / Each part is a half. What shape is each part? / [picture: shapes2d]	Optional, only if you are on time. A square folded corner to corner makes 2 halves that are each a triangle. The picture on the slide shows only an up-and-down fold line, so children imagine the corner fold or try it with a spare square. Take two answers and move on. This links to Build the Figure next lesson. Answer: a triangle. Children check by folding a square corner to corner.
33	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Is a quarter always smaller than a half? / Make a puzzle | Fold a secret fold. Ask: half or quarter? / Teach it back | Show a toy how to check for halves. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, a ruler, paper and your paper shapes. Pause or rewind any time.*	Gather your paper shapes, pencil, ruler and paper. Press play when ready.
4:00	Remember warm-up	*Recording: join in! Share the dots, name the shapes, and say your answers out loud before I do.*	Play along. Draw the dots, name the shapes, answer out loud.
10:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles with a sheet of paper. Guessing is fine. Press play for the next part.*	Pause. Fold a sheet of paper for both puzzles. A guess is fine.
14:00	Explore	*Recording: pause now. Fold your circle so the edges meet, open it, and lay one part on the other.*	Pause. Fold your circle in half, then again. Fold a long rectangle twice. Try a crooked fold.
26:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
36:00	Brain break	*Recording: stand up and join in! Make a whole turn, a half turn and a quarter turn, slowly.*	Stand up. Turn slowly: a half turn, a quarter turn, then sit down.
40:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.

70:00	Talk time	*Recording: pause. Tell a grown-up whether any 2 parts of a shape are halves, and test it with your circle.*	Pause. Explain your answer to someone at home and test it with your paper.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one shape you folded.*	Show a grown-up your workbook and one folded shape.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	4:00	3:12	3:41	ok
remember	6:00	4:56	5:51	ok
sneak	4:00	3:14	3:47	ok
explore	12:00	10:06	11:58	ok
mini	10:00	8:13	9:41	ok
break	4:00	3:14	3:50	ok
work	30:00	21:27	25:30	ok
talk	6:00	4:26	5:16	ok
exit	6:00	4:32	5:21	ok
look	3:00	2:23	2:48	ok
flex	5:00	0:26	0:30	ok
Total	90:00	66:12	78:13	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 7.5 min for page 1 and 8.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.